Solution Architecture

Date	2 October 2026
Team ID	6abca5b16f395c49751f124d
Project Name	LegalEase: AI-Powered Legal Document Generator
Maximum Marks	4 Marks

Architecture Overview

User Browser
↓
Streamlit Frontend
↓ HTTP POST /generate
FastAPI Backend + Pydantic Validation
↓
GeminiDocumentGenerator
↓
Google Gemini API
↓ generated draft
FastAPI Response
↓
Editable Preview / Session State
↓
Export Service → TXT | DOCX | PDF

Component Responsibilities

Component	Responsibility
Streamlit Frontend	Collect inputs, display disclaimer, show preview, support editing and downloads.

FastAPI Backend	Expose routes, validate requests, orchestrate AI generation and return response.
Gemini Service	Build document-specific prompt, call Gemini, validate non-empty response and handle transient failures.
Export Service	Convert latest edited content to TXT, DOCX and PDF.
Configuration	Load GEMINI_API_KEY, model and backend URL from environment.
Tests	Verify routes, validation, AI-service behavior and export generation.